UNIT 2 · INTRODUCTION AND STATE OF ART OF AI, NLP & POTENTIAL USE OF AI

CHAPTER 2.2

Natural Language Processing

Teaching computers to read, listen, and talk like humans

Text

Translate

Chat

Search

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CHAPTER ROADMAP

Learning Objectives

After studying this chapter, you will be able to:

01
Understand NLP
What Natural Language Processing is and why it matters to AI

02
Text Generation
How AI models like GPT create human-like text

03
Language Translation
How machines convert text between languages

04
Question Answering
How chatbots interpret and answer questions

05
Dialogue Systems
How assistants like Siri and Alexa hold conversations

06
Internet Searches
The different intents behind every search query

DOMAIN OF NLP

Human Language vs. Computer Language

H

Humans

- Communicate through speech and writing
- The brain interprets meaning constantly, even switching focus between two conversations at once
- If meaning is unclear, we simply ask for clarification
- Language is flexible — full of tone, emotion, and context

C

Computers

- Communicate through numbers and structured codes
- Every input must follow exact rules and syntax
- Even a small error can cause a command to be rejected
- Language must be error-free — machines cannot easily guess intent

DOMAIN OF NLP

What is Natural Language Processing?

Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand, interpret, process, and generate human language in a natural, meaningful way.

In simple words: NLP teaches machines to read, listen, and talk like humans.

It draws on linguistics, computer science, and machine learning to extract hidden information — opinions, emotions, facts — from **unstructured data** such as chats, tweets, reviews, and articles.

NLP UNLOCKS

- Text summarization
- Language translation
- Chatbots & assistants
- Speech recognition
- Sentiment analysis

DOMAIN OF NLP

Why is NLP Difficult?

Computers expect structured, rule-based input — but human language breaks the rules constantly.

1

Ambiguity

The same word or sentence can carry multiple meanings depending on context.

2

Sarcasm & Tone

Some rules are abstract — sarcasm, humor, and emotion rarely follow a formula.

3

Slang & Culture

Language shifts with region, culture, and time, unlike fixed programming syntax.

4

Simple Exceptions

Even easy rules (like adding ‘s’ for plurals) have countless irregular exceptions.

Text Generation — How It Works

Text generation is AI creating human-like text — a story, poem, article, or even code — using **language models** such as GPT (Generative Pre-trained Transformer) and Google's PaLM.

Input Text

Raw sentence or prompt fed into the system

AI Model

Deep neural network trained on vast text data predicts the next words

Generated Text

Coherent, contextually relevant output text

Trained on massive internet-scale text, these models learn grammar, style, and patterns — then predict the most likely next word to build coherent output.

Examples: GPT-3 (OpenAI) • LaMDA (Google) • AI Dungeon

Applications & Limitations

✓ Applications

- Content creation — articles, blogs, product descriptions
- Chatbots & virtual assistants — conversational replies
- Language translation — real-time conversions
- Summarization — condensing long text

! Limitations

- May lack deep context understanding
- Relies heavily on the quality of training data
- Struggles with rare or unseen scenarios
- Raises ethical issues — bias or misinformation

TEXT UNDERSTANDING

Text Understanding (NLU)

Natural Language Understanding goes beyond words to grasp meaning and context.

1

Semantic Analysis

Understanding the meaning of words and phrases

2

Syntactic Analysis

Analyzing the grammatical structure of sentences

3

Contextual Understanding

Interpreting meaning based on surrounding context

4

Sentiment Analysis

Determining the emotional tone of a text

5

Entity Recognition

Identifying people, places, and organizations

TEXT UNDERSTANDING

Where NLU Is Applied

01 **Search Engines**

Understanding search queries to return relevant results

02 **Chatbots**

Comprehending user input to provide accurate responses

03 **Sentiment Analysis Tools**

Analyzing customer reviews and social media posts

04 **Information Extraction**

Automatically pulling key facts out of documents

LANGUAGE TRANSLATION

Machine Translation — The Process

Machine translation uses AI and machine learning to automatically translate text or speech from one language to another — a key application of NLP for cross-linguistic communication.

1

Input Processing

Break down the source language into words, grammar, and context

2

Model Application

Neural networks (e.g. Google's Transformer models) interpret meaning and context

3

Output Generation

Produce text or speech in the target language, preserving meaning and tone

LANGUAGE TRANSLATION

SMT vs. NMT

Statistical Machine Translation (SMT)

Uses statistical models to learn the probabilities of word and phrase correspondences between languages.

Relies on probability tables built from large bilingual text corpora.

Neural Machine Translation (NMT)

Employs deep learning, particularly sequence-to-sequence models, to learn complex mappings between languages.

Produces more fluent, natural-sounding translations — today's dominant approach.

Applications of Machine Translation

1

Tools

Google Translate, Bing Translator, DeepL for instant text and web translation

2

Cross-Border Communication

Diplomacy, travel, and international cooperation

3

Localization

Websites, apps, and software adapted for global audiences

4

Business & Commerce

Translating contracts, emails, and marketing materials

5

Education & Learning

Multilingual resources that support language learning

QUESTION ANSWERING

Chatbots — How They Work

Question answering chatbots automatically respond to questions posed in natural language.

1. **Analyze the Question**
   Identify the key information and intent behind the user's question

2. **Search the Knowledge Base**
   Look through documents or data to find relevant information

3. **Generate a Response**
   Produce a clear, human-like answer for the user

Human → Question → **Chatbot** → Answer → Human

QUESTION ANSWERING

Chatbots in Everyday Life

1

Customer Service

Instant support and answers to customer queries

2

Education

Answering student questions and offering resources

3

Information Retrieval

Helping users find information quickly

4

Virtual Assistants

Assisting with tasks and answering questions

Example: many websites deploy chatbots to instantly answer frequently asked questions.

DIALOGUE SYSTEMS

Smart Assistants (Dialogue Systems)

Smart assistants use NLP to interpret voice or text requests, then act on your instructions.

1. Answer questions ("What's the weather?")
2. Set reminders and alarms
3. Play music
4. Control smart home devices
5. Make calls & send messages
6. Provide news or sports scores

Examples: Siri (Apple) • Google Assistant • Amazon Alexa • Microsoft Cortana

With speech recognition, assistants don't just detect your voice — they make sense of it, and can access your data to keep notes, place calls, and send messages for you.

INTERNET SEARCH

Internet Search, by the Numbers

A search engine takes a query and returns a ranked list of relevant web pages — the gateway to the world's information.

2.2 trillion

search queries processed every year, across all search engines

92.82%

of the search market share held by Google (as of April 2023)

68%

of all online experiences begin with a search

8.5 billion+

searches processed by Google alone, every single day

TYPES OF INTERNET SEARCH

Navigational & Informational Searches

1 Navigational Searches

Guide users directly to a specific, familiar website or page — the user already knows their destination and uses search as a shortcut.

Examples: LinkedIn • YouTube login • Amazon Prime

"YouTube login" is navigational — it leads users directly to YouTube's login page.

2 Informational Searches

Learning-oriented queries where the goal is to gain knowledge, explore a topic, or answer a question out of curiosity.

Examples: How to bake a cake • Symptoms of the common cold • Best places to visit in Italy

Transactional & Investigative Searches

3 Transactional Searches

Action-oriented queries where the intent is to perform a specific task — buying, subscribing, or downloading.

Examples: Buy iPhone 16 • Order pizza online • Book a flight to London

"Subscribe to Netflix" is transactional — it initiates a subscription process.

4 Investigative Searches

Focused on comparison, reviews, and exploration — often during the research phase before a purchase.

Examples: Top-rated web series • Compare Samsung Galaxy vs. Google Pixel • Reviews of electric cars

TYPES OF INTERNET SEARCH

Voice Searches

5

A growing trend where users speak naturally to voice assistants — these queries often resemble ordinary human conversation rather than typed keywords.

- *“OK Google, what's the weather today?”*
- *“Hey Siri, find me directions to the nearest coffee shop.”*
- *“Alexa, play music by Taylor Swift.”*

REMEMBER

A search can belong to more than one category at once.

“Best laptops” is both informational (learning about laptops) and investigative (comparing options) at the same time.

CHAPTER RECAP

Key Terms at a Glance

NLP
AI branch enabling computers to understand and process human language

Text Generation
AI producing human-like written content (e.g. GPT, PaLM)

Machine Translation
Automatically translating text or speech between languages

Q&A Chatbots
Systems that interpret questions and generate valid answers

Navigational Search
Queries that guide users to a specific site or page

Informational Search
Queries seeking knowledge on a topic

Transactional Search
Queries with intent to buy, subscribe, or download

Investigative Search
Queries comparing or researching options

Questions?

Let's discuss the exercise questions and explore how NLP shapes the tools we use every day.

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